

Condition Classification from Drug Reviews using NLP & Deep Learning

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Problem Statement

- Online drug reviews contain valuable firsthand insights into patient experiences, side effects, and treatment outcomes that are often overlooked in traditional clinical studies.
- Manual review and analysis of thousands of patient comments is time-consuming, prone to bias, and inconsistent across different reviewers, making it difficult to extract structured, actionable information.
- Objective: Develop an automated NLP-based system to classify patient reviews according to their reported medical condition (e.g., Birth Control, Depression, Pain, Anxiety) to enable faster, scalable, and more consistent analysis.

Dataset Overview

- Source: Drugs.com dataset (~160k reviews)
- Features: Drug name, review, condition, rating
- Focused Conditions: Birth Control, Depression, Pain, Anxiety

Unnamed: 0	drugName	condition	review	rating	date	usefulCount	
0	206461	Valsartan	Left Ventricular Dysfunction	"It has no side effect, I take it in combinati...	9.0	May 20, 2012	27
1	95260	Guanfacine	ADHD	"My son is halfway through his fourth week of ...	8.0	April 27, 2010	192
2	92703	Lybrel	Birth Control	"I used to take another oral contraceptive, wh...	5.0	December 14, 2009	17
3	138000	Ortho Evra	Birth Control	"This is my first time using any form of birth...	8.0	November 3, 2015	10
4	35696	Buprenorphine / naloxone	Opiate Dependence	"Suboxone has completely turned my life around...	9.0	November 27, 2016	37

Data Preprocessing

- HTML removal, punctuation stripping, lemmatization
- Tokenization and stop-word removal
- Train-test split (80/20) and label encoding

Google Colab Link:

<https://colab.research.google.com/drive/15y7GNxfC7eesbgK406lit7RWGCe7B0CX?usp=sharing>

Feature Engineering

- Bag-of-Words (CountVectorizer)
- TF-IDF with n-gram ranges (1,2,3)
- Word clouds for top tokens

ML Models

- Multinomial Naive Bayes
- Passive Aggressive Classifier
- Accuracy: 96.5% with TF-IDF (n-grams)

Deep Learning – LSTM

- Embedding + BiLSTM + Dense
- Sparse categorical cross entropy loss
- Padded sequences for uniform input length
- Accuracy: 94.79%

Key Insights

- Passive Aggressive Classifier with TF-IDF (trigrams) achieved 96.6% accuracy, outperforming other models.
- Bidirectional LSTM reached 94.7% accuracy, capturing deeper text patterns.
- N-gram features significantly boosted prediction of medical phrases like “severe side effects.”
- Passive Aggressive was faster and highly accurate, ideal for efficient deployment.
- LSTM handled complex, longer reviews with stronger sequence understanding.

Future Work

- Expand to sentiment analysis alongside condition classification.
- Fine-tune models with pre-trained embeddings (GloVe, Word2Vec).
- Explore transformer-based models (e.g., BERT) for improved generalization.
- Enable multi-label classification for reviews mentioning multiple conditions.
- Develop a real-time web app for instant review classification.

Thank you!

Any Questions?